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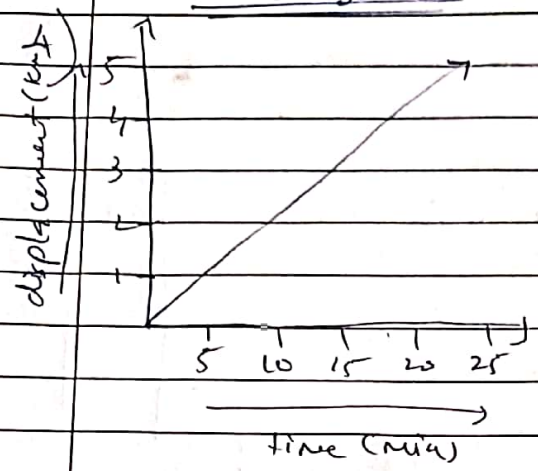
Exercise 2B



5) (i) You can calculate the acceleration from a velocity time graph by calculating the slope.

(ii) The Area ~~will give you~~ of the shape formed after drawing the line will give you the distance covered.

(iii) You can just directly mark a point on the time axis and draw a straight line to the displacement axis which will give you the value.

6) a) The body is at rest
b) Its going away from starting point
c) Its going towards the starting point.
d) motion with variable velocity.

7) Its because the slope of A is more than the slope of line B.

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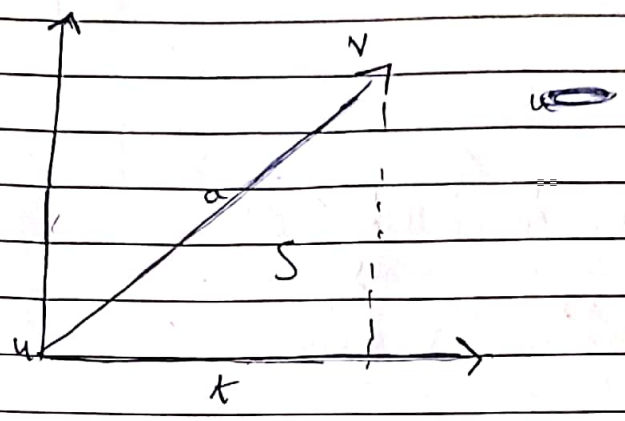
8) (a) Uniform Acceleration. Example: Free fall
(b) Variable Retardation. Example: A Feather falling down.

9)



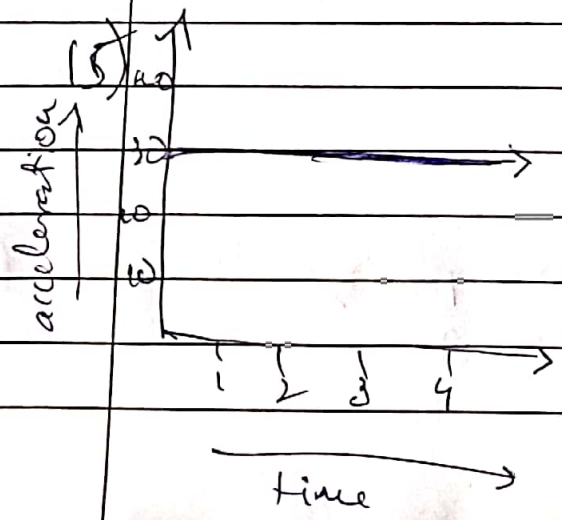
14) A: There is no acceleration because the velocity is constant and not increasing.

B: There is no acceleration because the slope is straight which means there is a constant velocity.

C: Negative acceleration because the slope is curving with time.

D: Positive acceleration because the slope is curving with time.

15)



You can find the area of the rectangle formed which will give you the final velocity. From the final velocity you can calculate the change in velocity.

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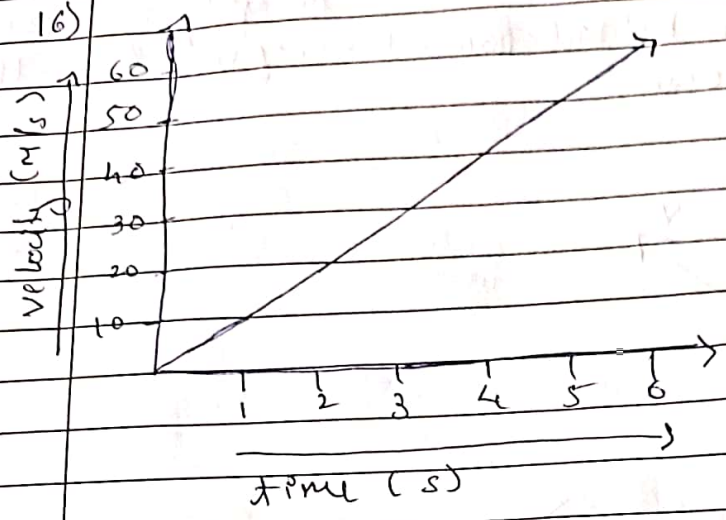
16)

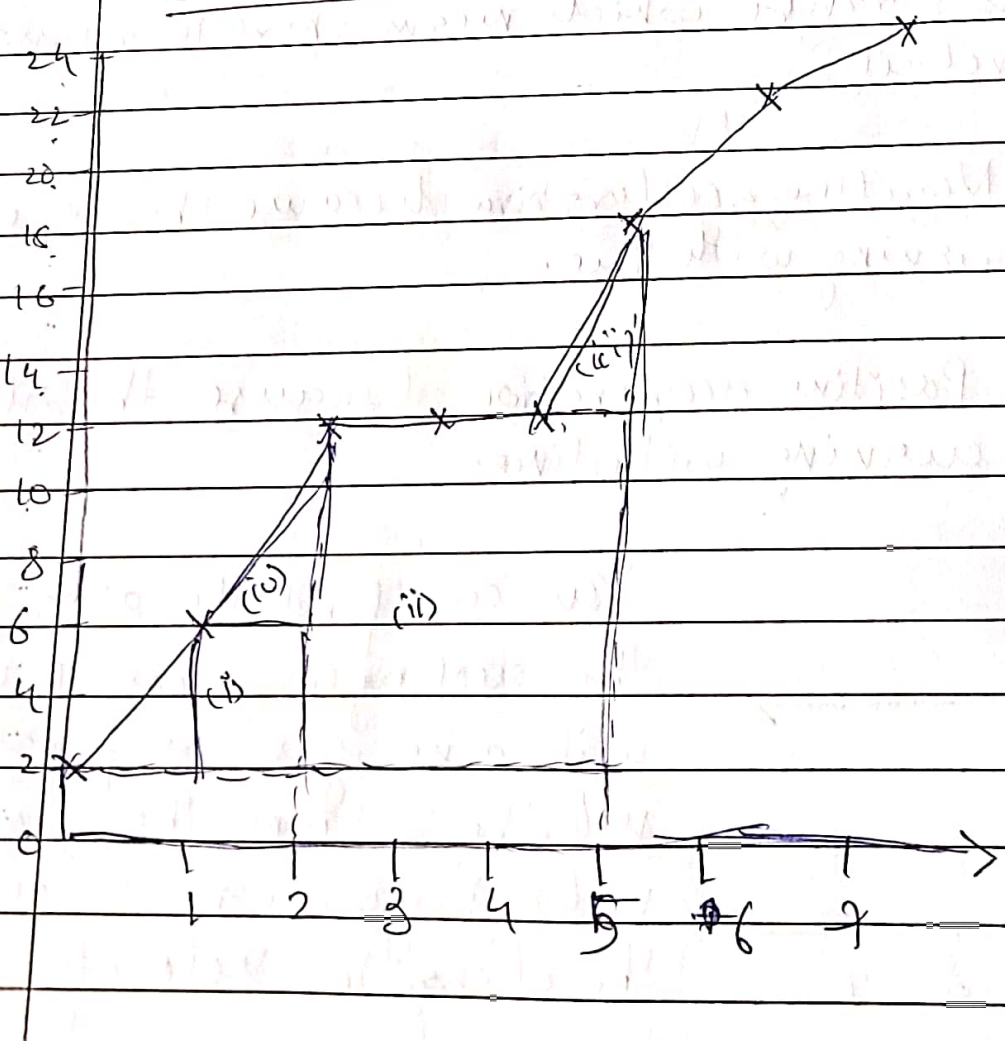


MCQs

1) (4) 2) (d) 3) (c)

Numericals :-

3)



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(i)

~~distance travelled $= \left(\frac{1}{2} \times 10 \times 2\right) + (10 \times 3) + \left(\frac{1}{2} \times 6 \times 1\right)$~~

~~$\Rightarrow 10 + 30 + 3$~~

~~$= 43\ m$~~

$dis = \left(\frac{1}{2} \times 6\right) + 4 + (10 \times 3) + 3$

$\Rightarrow 3 + 4 + 3 + 10 \times 3$

$\Rightarrow 40\ m$

(ii) avg velocity $= \frac{40}{4} = 10\ m/s$

4) a)

(i) velocity $= \frac{3}{5} = 0.6\ m/s$

(ii) velocity $= 0\ m/s$

(iii) velocity $= \frac{4}{2} = 2\ m/s$

(4) velocity $= \frac{4}{4} = 1\ m/s$

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5) (i) avg velocity $= \frac{10}{4} = 2.5$ m/s

(ii)
dis $= -10$ m

(iii) 7s and 13 s

6) (i) 40 km

(ii) A $= \frac{160}{4} = 40$ km/h

B $= \frac{80}{4} = 20$ km/h

(iii) 2 hours

(iv) 80 km

9) (i) By finding the area of the shape formed in the particular part

(ii)
Area of BC $= vt$
Area of AB $= \frac{1}{2} vt$

BC : AB :: 2 : 1

(iii) (a) BC
(b) AB
(c) CD

(iv) (a) It lower because it decelerates to D in less time.

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